AgRobot: Towards AI-Powered Crop Disease Detection and Medication Recommendation Robot

K. Yousef, M. Mohamed
Modern Science and Arts University, Egypt, yousef.ahmed8@msa.edu.eg, Mohamed.magdy28@msa.edu.eg

G. Tarek
MSA, Egypt, tghoniemy@msa.edu.eg

Abstract– Since Egypt's agriculture is deeply rooted in its rich history, with the Nile River playing a pivotal role, maintaining food security has always been important. However, plant infections pose a significant threat, reducing crop yields and impacting agricultural productivity. Traditional manual detection methods are insufficient, necessitating innovative solutions. Thus, this research proposes the development of an advanced inspection robot, "AgRobot," equipped with Deep Learning techniques to autonomously detect plant diseases. Leveraging the power of the NVIDIA Jetson Nano embedded system and Rapoo webcam, along with Yolov8 and Mobilenetv2 deep learning models, AgRobot aims to revolutionize disease detection in greenhouse farming, thereby contributing to enhanced food security and sustainable agricultural practices in Egypt. Moreover, various control strategies are explored, and fuzzy-PID controller shows the best navigation performance in terms of stability and efficiency due to its adaptive nature in parameter selection. The project has achieved an average detection accuracy of 96% with 20 frames per second (fps) on the Jetson Nano, demonstrating the feasibility and efficiency of the proposed solution.

Keywords-- Disease detection, AI-powered robot, greenhouse farming, yolov8, mobilenetv2

I. Introduction

The Agri-Bot project has the potential to transform disease management for both tomato and potato crops in Egypt, a nation where agriculture is vital. Agriculture contributes roughly 11% of Egypt's GDP and employs 20% of the workforce [1]. However, these key crops face numerous diseases. Agri-Bot empowers farmers to recognize common threats in tomatoes (early blight, late blight, fusarium wilt, verticillium wilt) and potatoes (potato early blight, late blight). This knowledge allows for comprehensive disease management strategies with preventative measures and treatment options. The project emphasizes consistent scouting and monitoring as crucial for disease prevention in both crops. By adopting a proactive, unified approach, Agri-Bot can empower Egyptian farmers to protect their crops, optimize yields, and ensure the continued success of this critical agricultural sector.

First thing to be considered is that plants' leaves are susceptible to various foliar diseases that can seriously affect their health and productivity. Traditional methods of disease detection require manual examination and expertise, which can lead to delays in treatment. These diseases affect not only the yield of plants, but also their quality. Additionally, farmers may experience losses in terms of income and resources for disease prevention, which can have a substantial impact on their financial situation. The issues are made worse by environmental effects including increasing pesticide use and probable soil contamination. To maintain a resilient and sustainable tomato and potato business, it is essential to recognize and understand the nature of these illnesses in order to develop effective preventative and treatment techniques.

The proposed method revolves around the "AgRobot," a mobile robot that combines sophisticated Artificial intelligence-based imaging capabilities with advanced data analysis equipped with high performance computing ARM--based embedded systems. The AgRobot swiftly analyzes images of tomato and potato leaves in near real-time, detecting diseases such as early blight, late blight, and bacterial spots with exceptional accuracy. Moreover, the Agri-Bot offers farmers precise medication recommendations, drawing from a wealth of disease-specific treatment options.

By seamlessly integrating technology into agriculture, this project seeks to empower farmers with accurate disease diagnosis and sustainable medication guidance, ultimately enhancing crop health, increasing yield, and promoting responsible farming practices on a global scale.

II. Literature review

Disease detection requires a cooperation between robots to produce a merged vision-based system [2]. The detection accuracy of a robot defined as the percentage of identified disease to the total number of infected crops. There are two types of cameras used each according to its functionality and configuration to give the suitable accuracy needed. the usage of better cameras does not give the ability to get higher accuracy, for example: utilizing a colour camera demonstrated good accuracy for the detection of tomato spotted wilt virus (TSWV) and powdery mildew in greenhouse pepper plants, whereas utilizing a multispectral camera produced accuracy for these two diseases of 80% and 61%, respectively. beside that the RGB cameras gives range of accuracy depending on the lighting conditions.

The recent technologies that is available in ground mobile robots. Ground mobile robots offer advantages over airborne platforms in terms of close-range data acquisition, weather independence, lack of legislative regulations, and higher payload. Robotic platforms, such as unmanned ground vehicles (UGVs), are designed for rough agricultural terrains

with using Different kinematic solutions. They are divided into three categories: Custom mobile robots, sensorized agricultural machines, and commercial solutions. Despite the variety in kinematic solution in UGV but, the wheeled skid-steered is the most used kinematic solution with a percentage of 44.0% beside there are many commercial platforms with different weights and dimensions. Moreover, the computational power is essential for building the platform there are a different class to choose between. The jetson nano is the most suitable computer to build the platform as it is a cheap board and have a small dimension [3].

AN AI-based pattern classification system that combined convolutional neural networks (CNNs) with support vector machines (SVMs) adopted with transfer learning is proposed in [4]. The proposed method explores an acceptable classification accuracy even with small dataset sizes by customizing pre-trained CNN models such as AlexNet, Vgg16, and ResNet for the purpose. However, the pre-trained model was chosen to support real-time processing, which may have limited its applicability to specific hardware configurations. In terms of classification performance and processing times, the suggested architecture which is based on the pre-trained AlexNet model outperforms better than other approaches, indicating its potential for real-time processing tasks.

A YOLOX-based technique for tomato leaf disease diagnosis, which improves accuracy and performance. To overcome sample imbalance, a unique confidence loss function is devised, and MobileNetV3 is used to provide lightweight feature extraction. The Convolutional Block Attention Module improves feature extraction even further. Data augmentation with CycleGAN corrects sample imbalances. When compared to the original model, the accuracy is improved by 1.27%, memory usage is reduced by 35.34%, and speed is increased by 50.20%. The model, quantized for the Jetson Nano device, operates at 11.1 FPS, providing an efficient option for tomato leaf disease identification, but with potential hardware limits [5].

TABLE I
COMPARISON OF MODEL PERFORMANCE ON JETSON NANO [5]

NETWORK MODEL	TIME SPENT IN BUILD PHASES	FPS
YOLOX	27.16	2.13
YOLOX-MOBILENETV3	22.83	3.57
YOLOX-MOBILENETV3-TENSORRT(FP32)	13.41	7.14
YOLOX-MOBILENETV3-TENSORRT(FP16)	5.43	11.11

A comparison between using the machine learning algorithm and deep learning algorithm in detecting the tomato leaf disease is proposed in [6]. The machine learning algorithms have faced three main challenges. firstly, the high cost to train model and implement the model in a mobile robot due to huge data implemented. Moreover, the lack ability of creating a generalized model and unequal model for disease detection. although, the slow speed of processing could cause adaption to a particular production and over fitted model. On a second view, the deep learning algorithms have the ability to perform the complex tasks as well as the ability to modify and optimize the model according to task parameter to give more efficiency and performance. furthermore, machine learning and deep learning have a variety accuracy of detection as mentioned in [6] as in R-CNN which gives accuracy up to 82.5 % which is better than machine learning method such as SSD by difference in accuracy by 13% and MLP by 17.61%.

The CNN analyses picture components over 4 layers and achieves 97-98% testing accuracy. SVM uses support vectors to generate a hyperplane to split data into classes while maximizing the margin. The Decision Tree algorithm organizes data by qualities and labels, employing the Gini index approach for binary splits. For leaf disease classification, the comparison analysis shows that CNN, SVM, Decision Tree, and K-Nearest Neighbor (K-NN) classifiers obtain accuracies of up to 97%, 97%, 90.3%, and 80%, respectively. The method is highly accurate, especially with CNN and SVM, although it may be limited by the processing resources required, notably for the CNN model [7].

[8] proposed a 13-layer Convolutional Neural Network (CNN) model with multiple optimizers adopted with a dataset from the Plant Village repository that contains 22,930 tomato plant leaf photos classified into ten types. This strategy explores a high disease diagnosis accuracy and the ability to test multiple optimizers. However, the hardware specs and resource requirements for training and testing were not explicitly stated, which may cause limits when scaling up or deploying the model on multiple platforms.

TABLE II
COMPARATIVE STUDY OF ACCURACY AND LOG LOSS OF THE MODEL WITH VARIOUS OPTIMIZER [8]

S.NO	OPTIMIZER USED	ACCURACY OBTAINED		LOSS OBTAINED	
		TRAIN	TEST	TRAIN	TEST
1.	Nadam	93.36	89.75	0.17	0.30
2.	Adam	95.35	93.78	0.13	0.09
3.	Adadelta	96.21	89.77	0.11	0.19
4.	Adamax	96.47	93.91	0.10	0.01
5.	RMSprop	89.92	87.85	0.36	0.14
6.	SGD	89.08	91.47	0.24	0.22

A Convolutional Neural Network (CNN) for disease diagnosis is proposed in [9]. With 643 image per class, the dataset is well-prepared, and data augmentation techniques are used. After 100 epochs, the model achieves a high accuracy of 99.02% in training and 99.01% in validation. The CNN has an accuracy of 99.98% and 99.99% in testing. While the Raspberry Pi may not be suited for initial model training due to its computing constraints, it offers a cost-effective choice for deploying the previously trained model on-site, allowing farmers to diagnose leaf diseases in a practical and economical manner. The research also offers possible enhancements to mobile device compatibility.

A comparative study for AlexNet and SqueezeNet CNNs is proposed in [10]. AlexNet, a pioneering CNN, was chosen for its great accuracy, while SqueezeNet was chosen for its small model size. A dataset compromises segmented tomato leaf pictures, and the hardware used for training and testing is

an Nvidia Jetson Tx1 running Ubuntu. AlexNet marginally beat SqueezeNet in terms of performance, although SqueezeNet had a substantially lower model size. Because of their lightweight and low processing needs, smaller networks like SqueezeNet are useful for mobile apps, making updates more efficient. This study also emphasizes the possibility for on-board training, which would allow for real-time illness diagnosis and further work on leaf extraction from complicated backgrounds.

TABLE III
COMPARISON BETWEEN ALEXNET, SQUEEZENET V1.1, MODEL IN [10]

NETWORK NAME	ALEXNET	SQUEEZENET	OURS
ACCURACY ON TEST SET	0.9565	0.943	0.96
MODEL SIZE	227.6MB	2.9MB	23.51MB
INFERENCE TIME	~150MS	~50MS	~650MS

A real-time automated system for evaluating the health of bean leaves using an advanced Deep neural network (DNNs) is presented in [11]. The U-Net and pre-trained ResNet34 models were used for leaf detection, while the Densenet121 model was used for health categorization. This technique has several advantages, such as high accuracy (98.31% classification accuracy rate in binary and multi-class tasks). However, the system's performance is dependent on hardware, like the camera-equipped robot, which may not be easily accessible to all farmers.

A robot with a simple mechanism, making it a cost-effective alternative for farming purposes is presented in a comprehensive study [12]. The study investigates the robot's navigation accuracy in a variety of circumstances, including straight and square courses, with excellent results obtained on stiff surfaces such as concrete and compacted earth. However, for softer soil found in greenhouses, it is claimed that compacting the robot's path can increase navigation accuracy. While the robot's design is simple and inexpensive, its effectiveness may be limited in some soil conditions, notably soft soil situations.

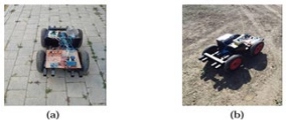

Fig. 1 Farming on Various surface a- Concrete; b- Soil [12]

A path planning strategy for non-holonomic mobile robots operating in a variety of situations is presented in [13]. Path planning is generally divided into two categories: global path planning, which is ideal for static settings with comprehensive previous information, and local path planning, which adjusts to dynamic impediments and the robot's kinematics in real-time. The research investigated path planning approaches such as RRT* and A* for global planning, Reeds-Shepp curve for local planning, and DWA as an alternative. According to the evaluation, the combination of A* and Reeds-Shepp curve-based local planning, aided by the E-band planner, obtains a success rate of 80% and creates shorter pathways than other techniques. The study emphasizes issues such as path feasibility, processing time, and the likelihood of creating a feasible path. Future study might include enhancing recovery behavior and evaluating navigation in uneven terrain situations while taking vibration costs into account.

Fig. 2 Mobile robot navigation in green-house [13]

A methodology for classifying tomato plant diseases from leaf images using MobileNet V2 is proposed in [14]. The trials were carried out on an HP Intel Core i7-3770 CPU with 8 GB memory, utilizing CPU-based TensorFlow in Python and no GPU. To improve efficiency, several hyper-parameter setups were investigated. The most effective results were achieved when Adagrad was used for training, with a batch size of 16, a learning rate of 0.001, and a data division ratio of 4:1 between training and testing sets. While this technology is accurate, its reliance on CPU-based processing may limit its speed and performance in resource-intensive applications.

[15] employed two deep learning models, Inception V3 and Inception ResNet V2, to detect tomato leaf diseases using a combination of laboratory and field image data. The methodology involved image preprocessing, data augmentation, transfer learning, and testing various dropout rates (5-50%) to optimize model performance. The best-performing models achieved an impressive 99.22% accuracy, surpassing several previous approaches in the field. Key advantages of this approach include its high accuracy and the incorporation of real-world field images alongside laboratory data, enhancing the models' practical applicability. While specific limitations were not explicitly stated in the summary, potential disadvantages of such deep learning approaches could include high computational requirements and the need for large, labeled datasets for training.

An AI-based system for detecting tea leaf diseases using the YOLOv7 object detection model. The researchers collected 4000 images of five common tea leaf diseases from tea gardens in Bangladesh, manually annotated them, and used data augmentation to expand the dataset. They trained the YOLOv7 model (dubbed YOLO-T) on this dataset and compared its performance to YOLOv5 and other existing methods. The YOLO-T model achieved high accuracy (97.3%), precision (96.7%), and recall (96.4%) in detecting the diseases. Key advantages were the model's ability to detect multiple diseases in natural scene images and its superior performance compared to previous methods. The main limitation was the long training time required. The study used a Canon EOS 80D camera for image acquisition and PyTorch for model implementation. While successful, the authors noted the need for a larger, more diverse dataset and potential improvements like batch normalization to reduce training time in future work [16].

A deep learning model called FPN-ISResNet-Faster RCNN to improve the accuracy of detecting diseases in apple tree leaves, particularly small and dense disease spots. The model combines a Feature Pyramid Network (FPN), an Inception Squeeze-and-Excitation ResNet (ISResNet), and Faster RCNN. It uses data augmentation techniques to expand

the training dataset and improve generalizability. The model achieves high accuracy (AP of 62.71%, AP50 of 93.68%, and AP75 of 70.94%) in detecting powdery mildew, Alternaria leaf spots, and rust disease. Key components include the ISResNet structure for feature extraction, FPN for multiscale feature learning, and optimized anchor frames. Advantages include improved detection of small disease spots and high accuracy across multiple diseases. Limitations include longer training time compared to simpler models and occasional misidentifications or missed the study suggests potential improvements such as online image enhancement and exploring different optimizers and loss functions to further enhance performance [17].

Fig. 3 Visualized detection results of Alternaria leaf spots [17]

[18] proposed an enhanced lightweight convolutional neural network model called GoogLeNet-EL for recognizing pepper leaf diseases. The methodology involves compressing the Inception module structure, introducing spatial pyramid pooling (SPP) to integrate multi-scale features, and optimizing components like using LeakyReLU activation, batch normalization without dropout, and the Adam optimizer. The model was trained and tested on a dataset of over 9,000 pepper leaf disease images. Key advantages include improved accuracy (97.87%) compared to other models, reduced memory requirements (10.3MB), and faster inference time. Limitations include that it was only tested on pepper diseases and may not generalize to other crops. The model outperformed alternatives like AlexNet, ResNet-50 and MobileNet-V2 in terms of accuracy and efficiency. However, it still requires further work to be deployed practically for farmers, such as integration with mobile apps. Overall, the lightweight design enables good performance on limited computing platforms, which is beneficial for potential field deployment, though more research is needed on practical applications.

III. Methodology

This section details the deep learning model employed for classifying plant diseases within agricultural images. This research paper leverage MobileNetV2, a lightweight and efficient convolutional neural network architecture specifically designed for image classification tasks on mobile and resource-constrained devices [19]. MobileNetV2 achieves this efficiency through innovative techniques like depth wise separable convolutions and inverted residuals, making it ideal for deployment in agricultural settings where computational resources might be limited [20]. The following subsections will delve deeper into the chosen dataset, the specific MobileNetV2 architecture used, the evaluation metrics employed to assess performance, and the classification results along with their discussion.

A. *Model Architecture*

The proposed model architecture is based on the MobileNetV2 network, pre-trained on the ImageNet dataset. The MobileNetV2 serves as the base model, which is loaded with pre-trained weights and excludes the top (fully connected) layers [21]. By freezing the base model layers, their pre-trained features are retained, and only the additional layers are trained to adapt the model to specific task. The model is extended by adding a Global Average Pooling to reduce spatial dimensions, followed by a dense layer with 512 units and Relu. Batch normalization is adopted, enhancing the model's generalization ability. A dropout layer with a rate of 0.5 is introduced to prevent overfitting. Subsequently, another dense layer with 256 units and Relu is added, followed by batch normalization. Finally, the output layer consists of a dense layer with a Softmax activation function, tailored to the number of classes in the dataset.

B. *Integrated pipeline*

The plant disease detection model pipelines two state-of-the-art deep learning models into one integrated workflow. First, the YOLOv8 object detector, identifies regions of interest on plants in the input image which are cropped and prepared for the MobileNetV2 image classifier. Finally, MobileNetV2 categorizes the plant regions into one of 13 disease class or healthy as figure 4. illustrates.

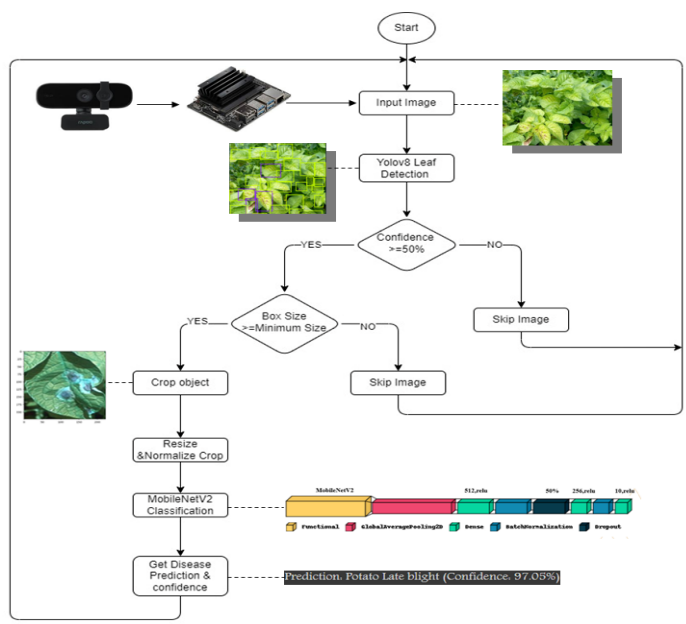

Fig. 4 Model pipeline flowchart

C. *Leaf detection model*

The leaf detection model in this study leverages YOLOv8 (You Only Look Once, version 8) for its object detection capabilities [22]. YOLOv8, known for its real-time efficiency and high accuracy, offers distinct advantages in agricultural applications. This algorithm excels at rapidly identifying leaves within images, a crucial aspect for immediate analysis

in the field. By drawing bounding boxes around each detected leaf instance, YOLOv8 facilitates precise localization, enabling subsequent tasks like disease detection or growth monitoring [23] This approach empowers the model to function effectively in real-world agricultural scenarios, ultimately contributing to informed decision-making for farmers.

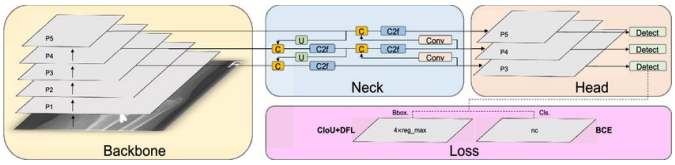

Fig. 5 The architecture of the YOLOv8 [24]

The architecture of YOLOv8, is a state-of-the-art object detection model designed to balance speed and accuracy for real-time tasks. It consists of three main components: the backbone, neck, and head as shown in figure 5. The backbone is responsible for feature extraction and employs advanced networks like CSPDarknet53 and EfficientDet to enhance learning capacity and efficiency. The neck, incorporating the Path Aggregation Network (PANet), connects the backbone to the head and facilitates multi-scale feature fusion, improving the model's ability to detect objects of varying sizes [24]. The YOLO head generates predictions, including bounding box coordinates, objectness scores, and class probabilities, based on the features from the backbone and neck.

IV. Results and Discussion

In this section, the paper delves into the results obtained from the training and evaluation of the proposed Deep Learning model designed to identify tomato disease images. The performance of the model is assessed through various metrics, including accuracy, validation accuracy, training loss, validation loss, ROC curve, and the confusion matrix. These metrics offer valuable insights into the model's ability to distinguish between tomato disease images.

A. Data Description

The dataset utilized for this study, titled "Plant Village" was acquired from Kaggle and comprises JPG images with a standardized size of 256x256 pixels. The images cover a range of tomato and potato leaf conditions, including 13 disease categories as shown in figure 6.

Fig. 6 Sample images from dataset

The dataset utilized in this research comprises 43,000 images, categorized into 13 distinct classes, as illustrated in Figure 7. To ensure comprehensive training and evaluation of the deep learning models, the dataset is systematically divided into three subsets: the training set, the validation set, and the testing set.

Specifically, the training set consists of 30,100 images, providing a robust foundation for the models to learn and generalize from diverse examples of plant diseases. The validation set includes 8,600 images, serving as a critical intermediary to fine-tune model parameters and prevent overfitting by assessing performance during the training process. Finally, the testing set is composed of 4,300 images, reserved for the ultimate evaluation of the model's accuracy and efficacy in real-world scenarios.

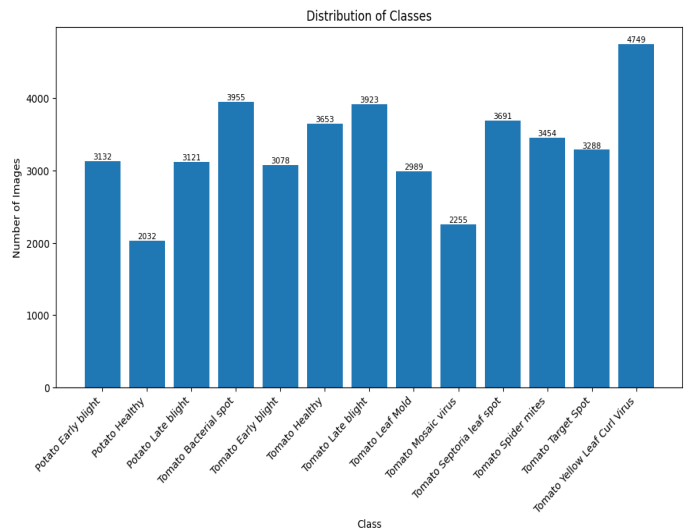

Fig. 7 Distribution of dataset classes

B. Data Annotation

To train the object detection model for the agricultural domain, a dataset of “plant village” is manually tagged with bounding boxes around the plant regions of interest. Figure 8. shows a sample of annotating process after labeling of 3000 images using Roboflow software (Yellow boxes for healthy leaves and blue boxes for infected leaves).

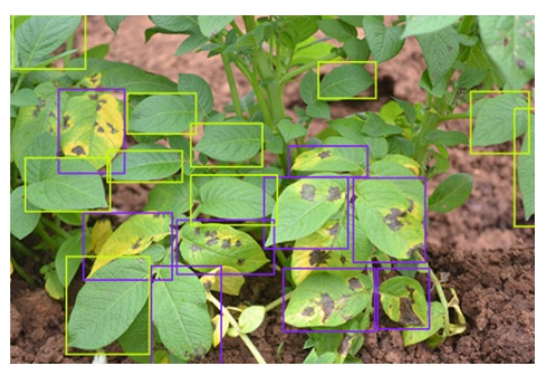

Fig. 8 Image Annotation sample

After annotating and labelling the data we spitted our data into train, valid and test set as shown in figure 9.

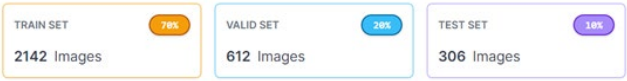

Fig. 9 Data splitting distribution

C. Objective Evaluation

In Figure 10.a, the proposed Deep Learning model achieved an 97% accuracy on the training set and 96% on the validation set for tomato and potato disease images. The accompanying training loss curve shows a steady decrease, indicating effective learning. Despite a slightly higher validation loss, the model did not overfit shown in Figure 10.b. These results highlight the model's robust generalization capabilities in accurately classifying unseen tomato and potato diseased image patches.

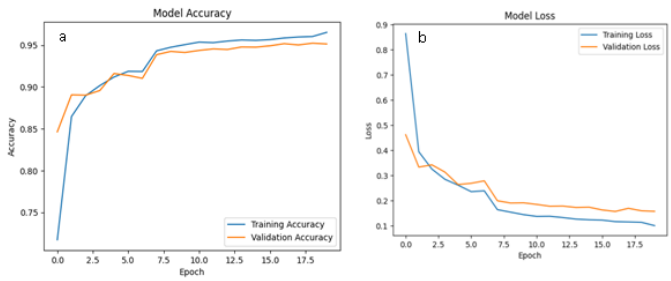

Fig. 10 (**a**) Training model accuracy; (**b**) Training model loss.

After training of the proposed Deep Learning model, the research paper conducted thorough testing to evaluate its performance in identifying plant disease images. The model demonstrated commendable effectiveness, achieving an overall accuracy of 91%.

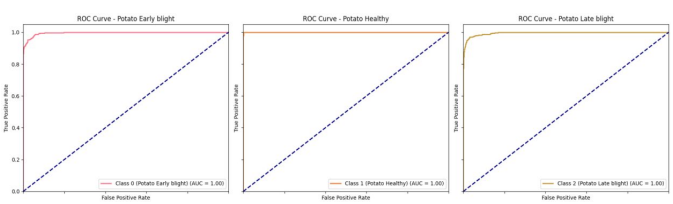

Fig. 11 Image Annotation sample

As illustrated in figure 11, the ROC curve explores the trade-off between the true positive rate and the false positive rate at different classification thresholds. A steeper ROC curve generally indicates a better-performing model. The proposed model achieved beyond all classes an area under the curve (AUC) of 0.89 - 1, indicating its robust performance in distinguishing different diseased images.

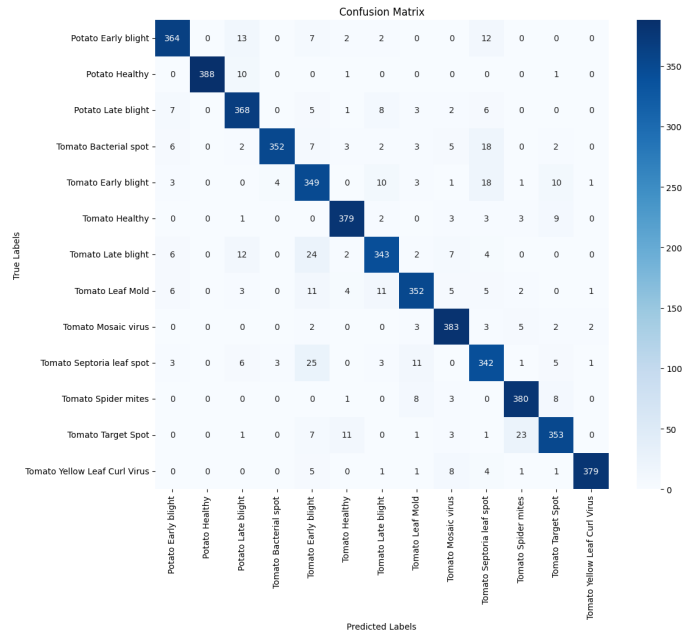

Fig. 12 Confusion matrix

Figure 13. below shows the predicted class names provides a visual representation of the model's performance. This visual assessment is crucial for understanding how well the deep learning model generalizes to real-world data. we can gain insights into the model's accuracy in classifying different types of plant diseases.

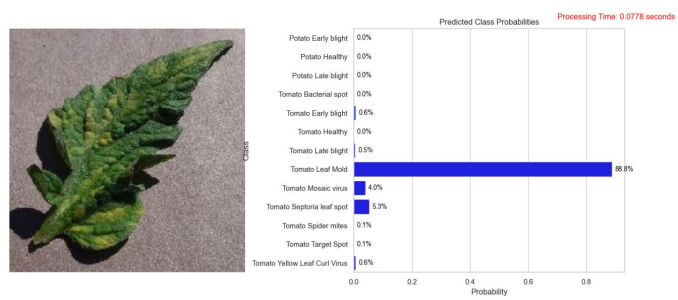

Fig. 13 Tomato Septoria leaf spot model classification result

In conclusion, the current plant disease classification model achieves an accuracy of 91%. Despite this success, the classifier exhibits inconsistencies when processing images containing multiple leaves, resulting in random classifications.

The efficiency of the fine-tuned YOLOv8 model for plant detection was tested using the usual COCO metrics.

- mAP50 (mean average precision at 50% IoU threshold): 0.98
- mAP50-95 (mAP averaged throughout IoU 0.5–0.95): 0.89

These scores represent the model's ability to properly detect plant items in photos while minimizing false positives as shown in figure 14.

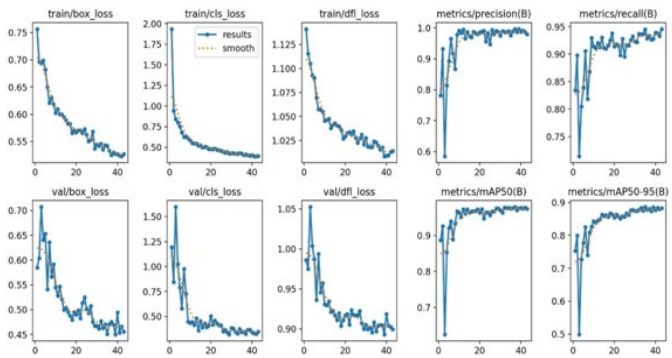

Fig. 14 Evaluation metrics

The fine-tuned YOLOv8 model improved the object detection performance on the test set as illustrated in. Figure 15.

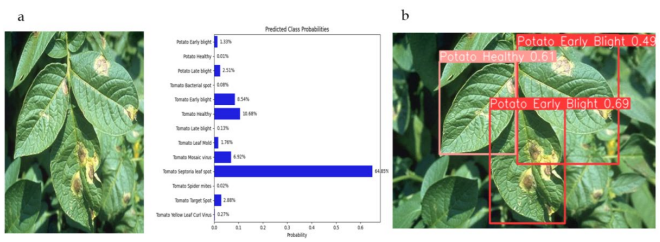

Fig. 15 a) Predicted before yolov8; b) Prediction after using yolov8

D. AgRobot Hardware Controller

The skid-steer mobile robot is a versatile platform widely used in various applications due to its maneuverability and ability to navigate challenging terrains [25]. It consists of a rigid chassis equipped with four wheels, typically arranged in two pairs, one on each side of the body as shown in figure 16.

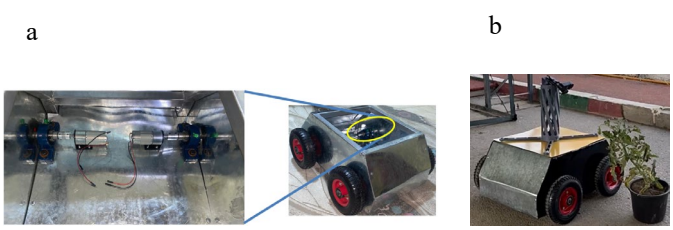

Fig. 16 a) 4-wheel skid steer mobile robot motor alignment; b) finalized robot

The following is the implementation of both PID and fuzzy PID controllers to govern the linear velocity of the mobile robot, we present the results in Figure 17. This figure encapsulates the comparative performance of the two control strategies, showcasing key metrics such as response time, stability, and error minimization. Through rigorous analysis, this paper aims to discern the most effective control approach for optimizing the robot's navigation and maneuverability in agricultural environments.

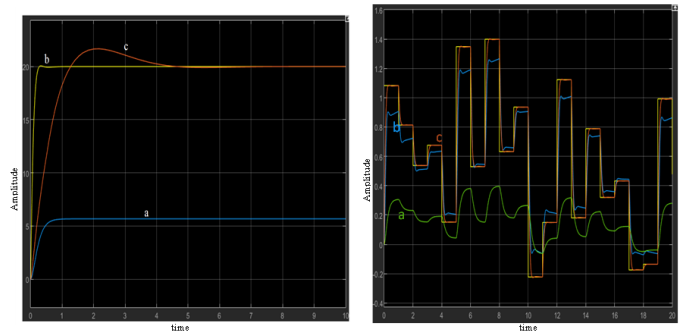

Fig. 17 a) Response without controller, b) With PID, c) With FUZZY-PID using step & random input

Table IV. shows a comparison between system response without control, with PID, and FUZZY-PID controllers: the FUZZY-PID controller demonstrates a better performance due its adaptive nature and broader range of operation. Unlike traditional PID controllers, which rely on fixed parameters and can struggle with varying system dynamics, the FUZZY-PID controller adjusts its parameters in real-time based on the system's current state. This adaptability allows it to handle a wider range of operating conditions effectively, resulting in more robust and efficient control.

TABLE IV

SYSTEM RESPONSE WITHOUT, WITH PID, FUZZY-PID CONTROLLERS COMPARISON

PARAMETER'S	WITHOUT CONTROLLER	PID	FUZZY - PID
OVERSHOOT (%)	0	0.1	4
RISE TIME (S)	1	0.5	4.5
STEADY STATE ERROR	14	0	0
SETTLING TIME (S)	0.4	0.1	0.8

E. AgRobot User Interface

The interface of the AgRobot plant disease detection system features circular indicators representing "Plant 1" and "Plant 2", dynamically linked to the actual number of plants in the greenhouse. These circles serve as visual representations of the health status of the monitored plants. Depending on the number of real plants in the greenhouse, the corresponding number of circles is displayed on the interface. Each circle changes color, either to green or red, reflecting the health condition of the respective plant it represents. This dynamic adaptation ensures that the interface accurately reflects the current state of the greenhouse, providing users with real-time insights into the health of their plants and enabling prompt action in case of any detected diseases or issues.

Fig. 18 a) Two plants diseased; b) Plant 1 is diseased and plant 2 is healthy

After plant selection, the interface displays the prediction of the plant's health status along with recommended treatment report. This information is fetched from the AgRobot plant disease detection system, providing users with valuable insights into the specific issues affecting their plants. The prediction text informs users about the detected condition, while the provided PDF file offers detailed guidance on the recommended treatment. This comprehensive approach empowers users to take informed actions to address any identified diseases or issues promptly, ultimately contributing to the overall health and productivity of their greenhouse. The recommendation section in user interface represents the recommended treatment instructions in both Arabic and English languages beside information about the disease causes. Figure 19 shows the treatment recommendation reports.

Fig. 19 Treatment recommendation (a) Arabic version (b) English version

V. CONCLUSION

This paper presents AgRobot, an autonomous mobile robot designed for early disease detection in crops using advanced deep learning. AgRobot combines YOLOv8 for accurate leaf detection with MobileNetV2 for robust disease classification. Through comprehensive testing, AgRobot achieved a remarkable 96% accuracy in detecting and classifying diseased leaves, laying the groundwork for proactive disease management in agriculture. Additionally, AgRobot operates in real-time on the Jetson Nano embedded platform, achieving 20 frames per second (fps). Furthermore, extensive exploration of control strategies revealed that a fuzzy PID controller offered superior navigation performance compared to others in terms of stability and efficiency. This finding led to its integration into the final design. Overall, this research demonstrates the potential of AI-driven mobile robots to revolutionize early disease detection in agriculture, ultimately promoting improved crop health and productivity.

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